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A. Qualitative analysis on CoT paths elicited by CoT-decoding and existing approaches

In Table 8, we present qualitative examples illustrating the distinctions in the generated CoTs for each method. Overall we observe that CoT-decoding exhibits a more "free-form" CoT generation in comparison to alternative CoT prompting methods. This divergence may be attributed to two factors: (1) we encourage the diversity at the initial decoding step, and (2) the absence of explicit constraints imposed by prompting.

Another noteworthy observation is that CoT-decoding can better reveal what LLMs' intrinsic strategy in solving a problem, without being influenced by the external prompts which could be biased by the prompt designers. Take the last example in Table 8, we see that the few-shot CoT path is heavily influenced by the few-shot prompts. Specifically, the few-shot prompts, sourced from (Suzgun et al., 2022), consistently follow a standard analytical approach – first assessing the person's profession, followed by an evaluation of whether the profession aligns with the action. This aligns with the standard method of solving this particular task.[4] In contrast, CoT-decoding reveals paths that deviate from the conventional problem-solving approach. Despite yielding an incorrect final answer according to the ground truth in some cases, the CoT paths remain to be valid.

[GSM8K] *Claire makes a 3 egg omelet every morning for breakfast. How many dozens of eggs will she eat in 4 weeks?* **Greedy**: 3 **CoT-decoding**: A dozen is 12. 3 eggs per day for 4 weeks is 84 eggs. 84/12 = 7. 7 dozen eggs. **Few-shot CoT**: There are 7 days in a week. 4 weeks is 4 * 7 = 28 days. She eats 3 eggs each day, so in 28 days she eats 28 * 3 = 84 eggs. 1 dozen is 12 eggs. 84 eggs is 84 / 12 = 7 dozen eggs. The answer is 7. **Zero-shot CoT**: First, how many eggs does she eat in a week? 7 days in a week, 3 eggs per day, so 7 x 3 = 21 eggs per week. Now, how many eggs does she eat in 4 weeks? 4 x 21 = 84 eggs. Now, how many dozens of eggs is that? 84 / 12 = 7 dozens.
[Year Parity] *Was Nicolas Cage born in an even or odd year?* **Greedy**: Nicolas Cage was born in an odd year. **CoT-decoding**: Cage was born in 1964, an even year. **Zero-shot CoT**: 1. Nicolas Cage was born in 1964. 2. 1964 is an even number. 3. Therefore, Nicolas Cage was born in an even year.
[Sports Understanding] *Is the following sentence plausible? "Walker Buehler earned a trip to the penalty box."* **Greedy**: No. **CoT-decoding** (the top few paths): - Sure, if he's playing hockey. - In the context of a baseball game, no. - I'm not sure what you mean by "plausible". It's a perfectly grammatical sentence, and it's a perfectly reasonable thing to say. **Few-shot CoT**: Walker Buehler is a baseball player. Penalty box is part of ice hockey. So the answer is no.

Table 8 | Example of generated CoTs using different approaches.

In Table 9, we illustrate with a concrete example of how the path aggregation algorithm works and why it could identify the correct answer more reliably. For the correct answer 18, we see it appears 4 times, each with $\Delta = 0.994$ ($k = 0$), $\Delta = 0.911$ ($k = 6$), $\Delta = 0.584$ ($k = 8$), and $\Delta = 0.999$ ($k = 9$), so overall its $\sum_k \Delta_{k,a} = 3.5$; while all other incorrect answer options $14, 16, 20, 10$ have a much lower sum of Δ.

[4] https://github.com/google/BIG-bench/tree/main/bigbench/benchmark_tasks/sports_understanding